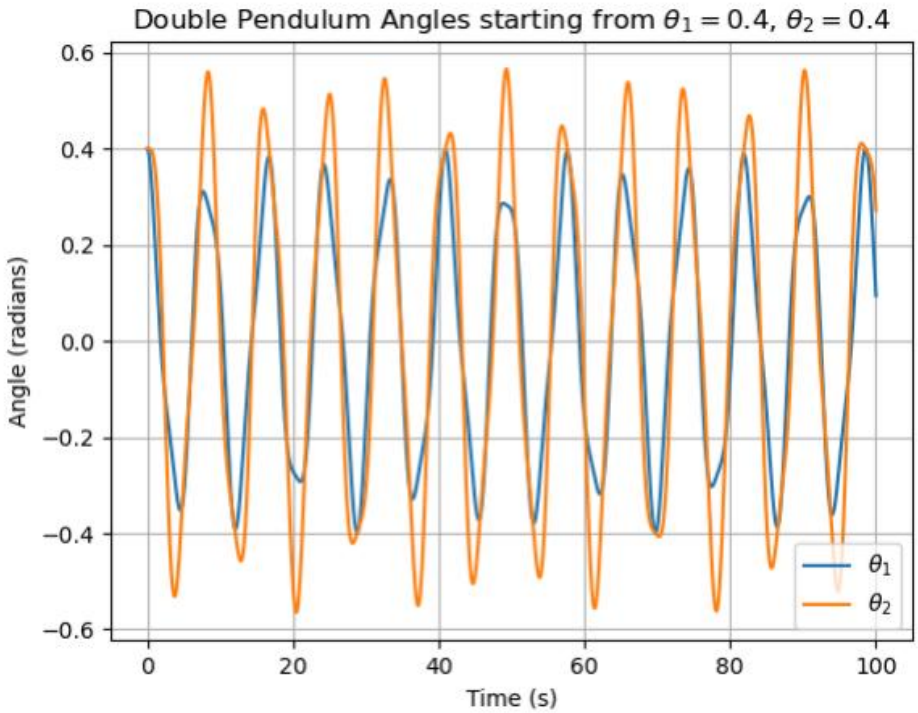
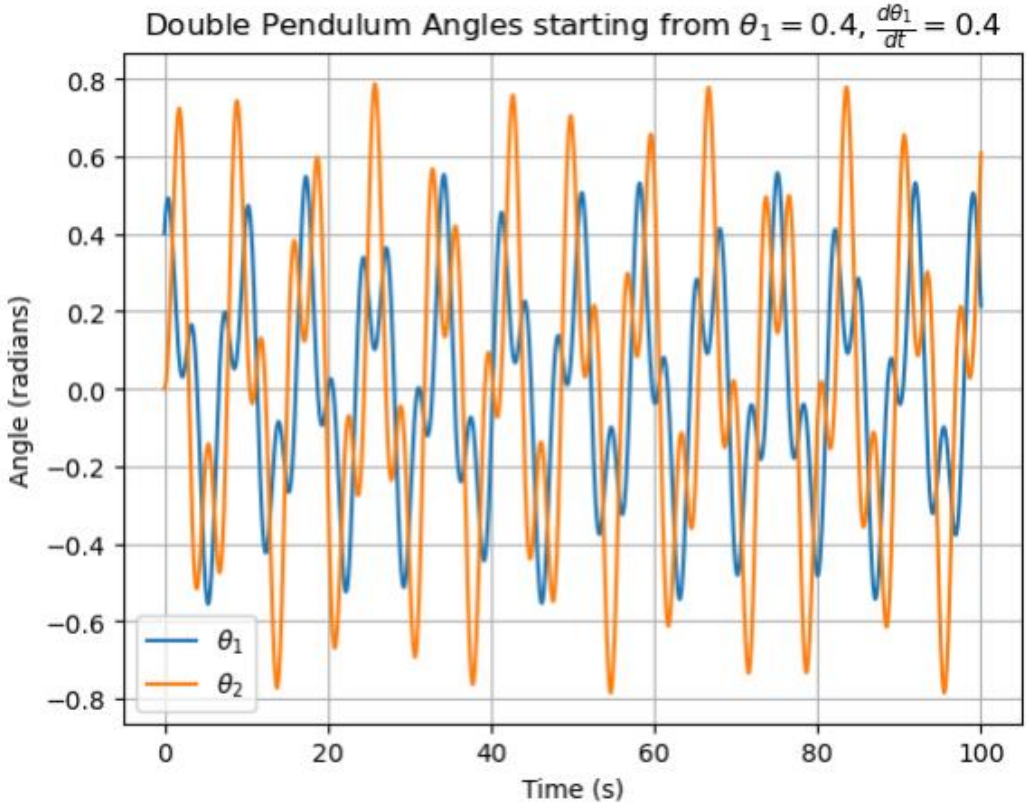
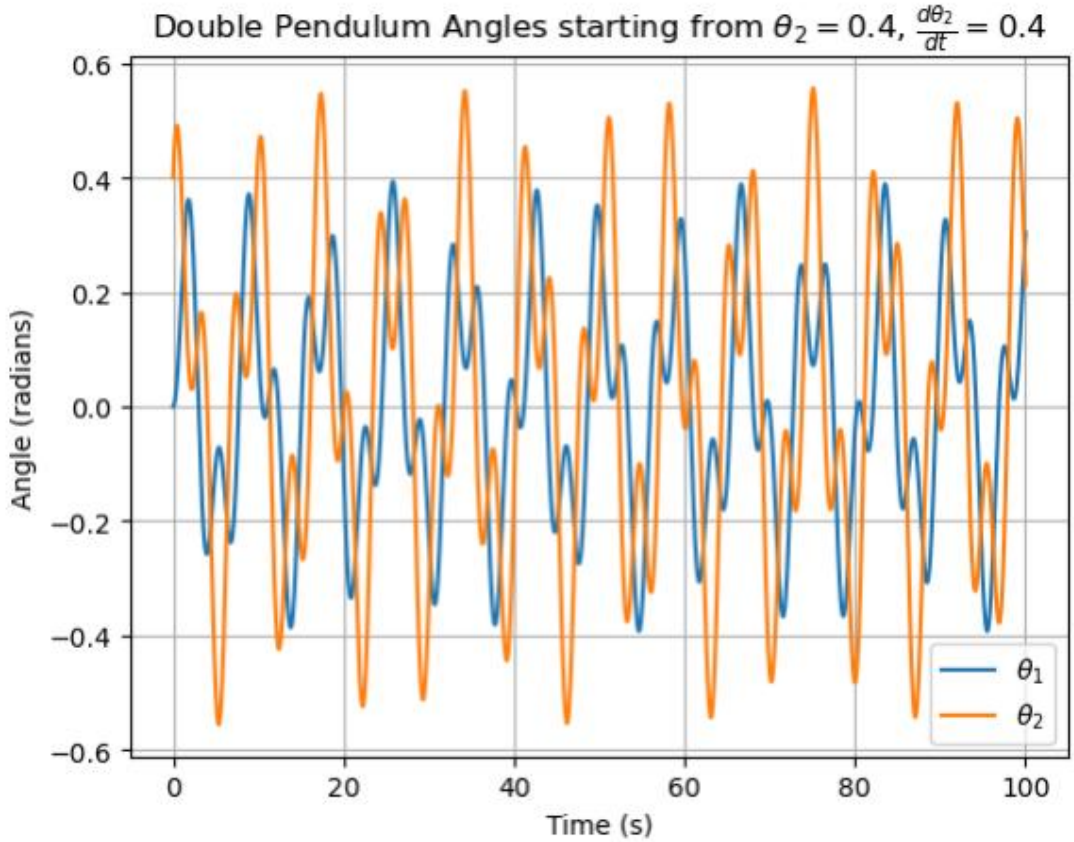
4b)

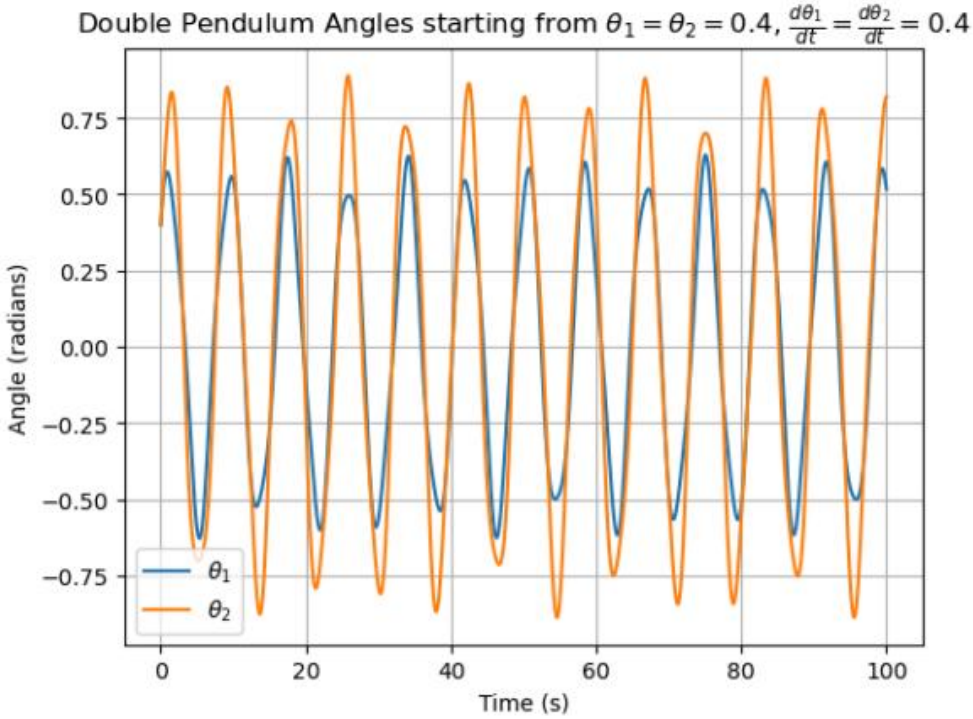
$$\frac{d}{dt}\underbrace{\begin{bmatrix}\theta_1\\ \theta_2\\ \dot\theta_1\\ \dot\theta_2\end{bmatrix}}_{x} = \underbrace{\begin{bmatrix}0 & 0 & 1 & 0\\ 0 & 0 & 0 & 1\\ -2 & 1 & 0 & 0\\ 2 & -2 & 0 & 0\end{bmatrix}}_{A}\underbrace{\begin{bmatrix}\theta_1\\ \theta_2\\ \dot\theta_1\\ \dot\theta_2\end{bmatrix}}_{x}$$

$$\frac{d}{dt}x = A\,x$$




Double Pendulum Angles starting from $\theta_1 = 0.4$, $\frac{d\theta_1}{dt} = 0.4$
Angle (radians)
Time (s)
θ_1
θ_2


Double Pendulum Angles starting from $\theta_2 = 0.4$, $\frac{d\theta_2}{dt} = 0.4$
Angle (radians)
Time (s)
θ_1
θ_2



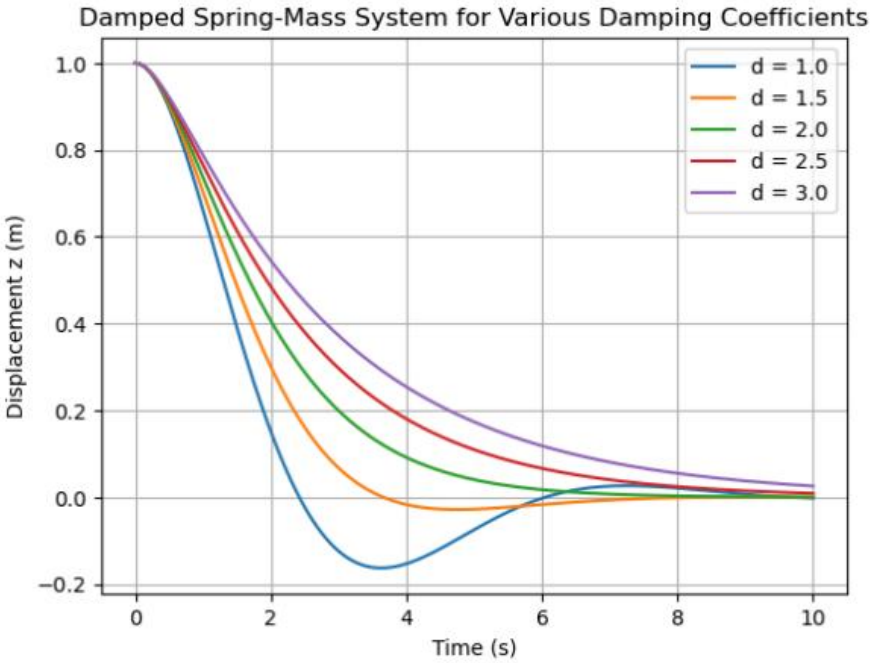
For the code, please refer to the appendix.

5. $m\ddot{z} + d\dot{z} + kz = 0 \rightarrow \ddot{z} = -\frac{d}{m}\dot{z} - \frac{k}{m}z$

a) State variables : $z, \dot{z}$

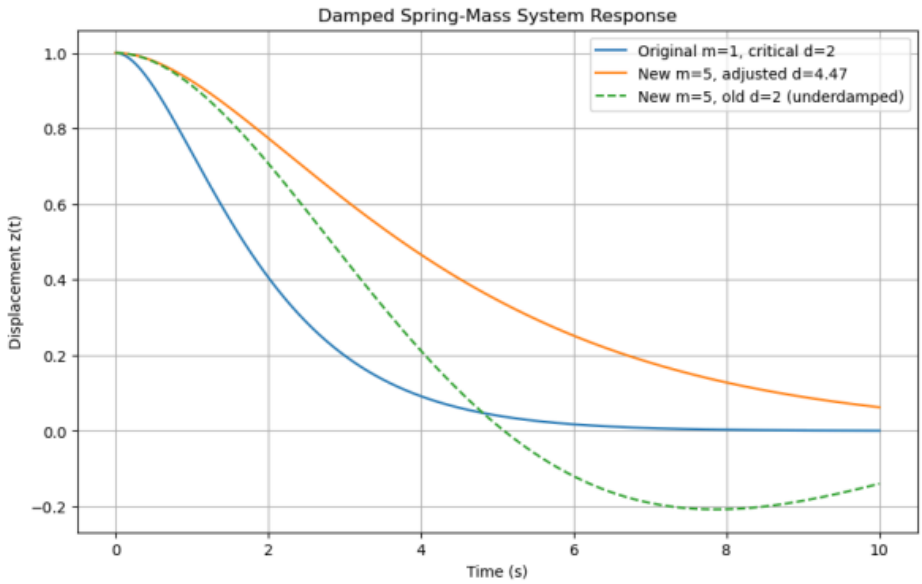
b) $\dot{x} = Ax$ format:

$$\frac{d}{dt}\underbrace{\begin{bmatrix} z \\ \dot{z} \end{bmatrix}}_{\vec{x}} = \begin{bmatrix} \dot{z} \\ \ddot{z} \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{d}{m} \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} z \\ \dot{z} \end{bmatrix}}_{\vec{x}}$$

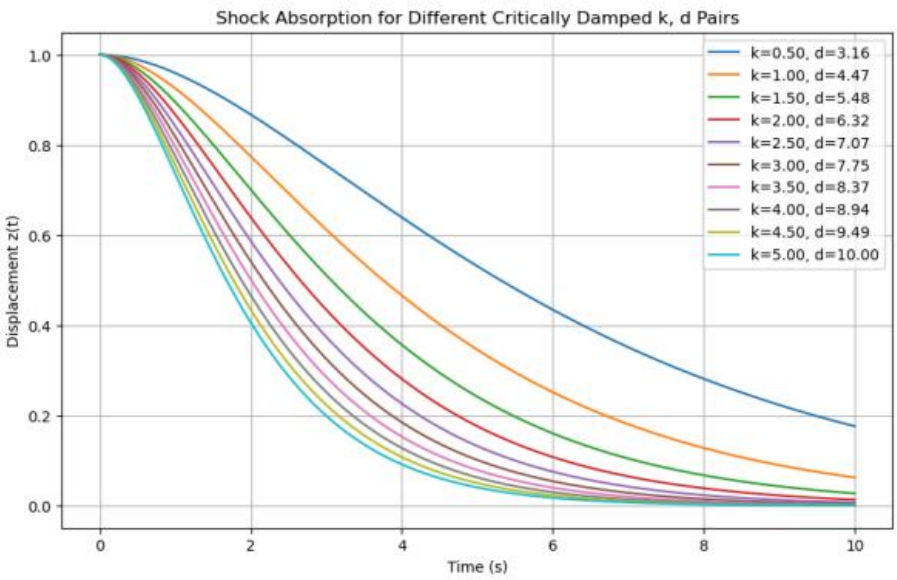


At $d = 2\sqrt{mk}$, $d = 2$ since $m = k = 1$. So, we are considering the green line. At critical damping, the system returns to equilibrium (displacement $z = 0$) as quickly as possible without any overshooting; as seen by the yellow line ($d = 1.5$), when the system is underdamped (damping less than the critical damping), there is quick return towards equilibrium, but the system overshoots and oscilates around equilibrium before settling. An overdamped system ($d > 2$, red and purple lines) will return to the equilibrium without oscillation but has a slower return to equilibrium than the critically damped system.

To understand why critical damping is ideal, consider a car's shock absorber. If the system is underdamped, the car bounces up and down (oscillates vertically) after hitting a bump. If the system is overdamped, the car slowly returns to its normal/equilibrium position, and may feel "heavy","sticky", or "sluggish." Critical damping provides the optimal balance, where the car quickly stabilizes without bouncing. Critical damping provides the fastest non-oscillatory return to a system's equilibrium.



$d_{crit} = 2\sqrt{mk}$. For critical damping, $d_{new} = 2\sqrt{5k_{new}}$.



As k increases, d also increases to maintain critical damping. Smaller spring constants k with smaller damping coefficients d lead to slower responses while larger k, d pairs lead to stiffer springs/heavier damping and faster responses. All critically damped systems have no oscilation. The "best" system depends on the application: for comfort, choose a small k and small d, but for fast response, choose large k, d pairs.

For code, please refer to appendix.

6.

a) $\ddot{z} = -z$; state variable: $z, \dot{z}$

$$\frac{d}{dt}\begin{bmatrix} z \\ \dot{z} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}\begin{bmatrix} z \\ \dot{z} \end{bmatrix}$$

$$\frac{d}{dt}\underbrace{\vec{z}}_{} = \underbrace{A}_{} \underbrace{\vec{z}}_{}$$

Alternate: any linear invertible transform of $(z, \dot{z})$ also works. e.g.:

$$\frac{d}{dt}\begin{bmatrix} z+\dot{z} \\ \dot{z} \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} z \\ \dot{z} \end{bmatrix}$$